

**Subject:**  
Behavioral Sciences

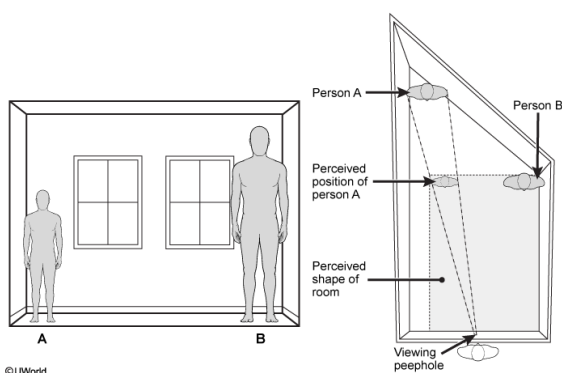
**System:**  
Sensation, Perception, and Consciousness

Visual perception involves the organization and interpretation of information obtained from visible light. Visual perception relies on monocular and binocular cues, as well as *perceptual organization principles* that allow humans to interpret limited or partial information more holistically. These sorts of mental shortcuts are based on prior experiences and knowledge, and are especially useful when lighting or perspective is limited. For example, the ambiguous image in Figure 1 is often perceived as a cup.



**Figure 1** Ambiguous image

Perceptual organization principles allow our brains to quickly and efficiently make assumptions about visual inputs, but these principles can also give rise to perceptual illusions. For example, the Ames room is constructed to exploit monocular depth cues in such a way that the person standing on one side of the room appears to impossibly dwarf the person on the other side of the room (Figure 2).



**Figure 2** Ames room illusion (two people standing in a specially constructed Ames room, viewed through a monocular peephole)

Human senses were adapted for use on land, which can lead to potentially deadly perceptual distortions while flying or underwater diving. Pilots are prone to such perceptual distortions when landing on unfamiliar runways. During their training, pilots develop a mental image of how a typical runway appears as they descend and land. Later, they

compare this prototypical shape to the runway they see before them to make adjustments

in the slope of their descent. However, if a runway differs from what the pilot is used to (eg, in width or slope), it will differ from the pilot's mental image for a typical runway at that same altitude. For example, an unusually wide runway will appear to be closer than it actually is, and a narrow runway will seem to be further away. This can cause the pilot to approach at a dangerously steep or shallow slope.

Pilots and divers can also suffer from spatial disorientation due to variations in sensory stimuli that are not typically encountered on land. For example, gravity is not experienced in the body the same way underwater as on land, and visual cues are often limited in deep dives. In these murky conditions, divers can easily lose track of the surface of the water, leading to extreme spatial disorientation and panic.

Spatial disorientation experienced by pilots and divers would be LEAST affected by information processed by the:

- ✓ ☒ A. organ of Corti. [44%]  
☐ B. otolith organs. [18%]  
☐ C. semicircular canals. [8%]  
☐ D. somatosensory system. [28%]

Correct

44% answered correctly

### Explanation:

| Bodily senses for orientation & movement                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kinesthetic sense                                                                                                                                                                                                                                                          | Vestibular sense                                                                                                                                                                                                                                                                                |
| <ul style="list-style-type: none"> <li>• Proprioceptors in muscles, joints, tendons &amp; skin detect movement &amp; position of body/limbs</li> <li>• Kinesthetic input provides awareness of location of parts of the body in space &amp; how they are moving</li> </ul> | <ul style="list-style-type: none"> <li>• Vestibular system maintains balance &amp; orientation, detects gravity/acceleration (otolith organs) &amp; head rotation (semicircular canals)</li> <li>• Motion sickness arises from conflict between vestibular system &amp; visual input</li> </ul> |

Maintenance of **spatial orientation** and **balance** relies on coordinated input from the visual system, the vestibular system, and the somatosensory system:

- **Visual input:** Visual cues provide information about the relative orientation of our bodies, objects, and light within our surroundings.
- **Vestibular input:** The vestibular organ within the inner ear provides information about motion, equilibrium, and spatial orientation. The organ is composed of three perpendicular **semicircular canals** and the **otolith organs** (utricle and saccule). The semicircular canals detect angular acceleration, whereas the otoliths detect linear acceleration (**Choices B and C**).
- **Somatosensory input:** Proprioceptors, found in skeletal muscles, tendons, skin, and

joints, provide information about the location and movement of our body in space  
(Choice D).

The brainstem, the cerebral cortex, and the **cerebellum** integrate these inputs to coordinate posture and balance, particularly via eye movements and postural adjustments.

The organ of Corti is located within the cochlea of the inner ear. Unlike the vestibular organs, the cochlea is *not directly involved in processing spatial information*. Instead, the main role of the cochlea is sound processing and transmission of auditory information to the temporal lobe of the brain.

**Educational objective:**

Spatial orientation and balance rely on input from the visual system, vestibular system (semicircular canals and otolith organs), and somatosensory system (proprioception).

Sensory input is processed by the brainstem and cerebral cortex, and by the cerebellum, which coordinates balance.

**Time spent:**0 sec

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Behavioral Sciences

**QID:**400281

**System:**

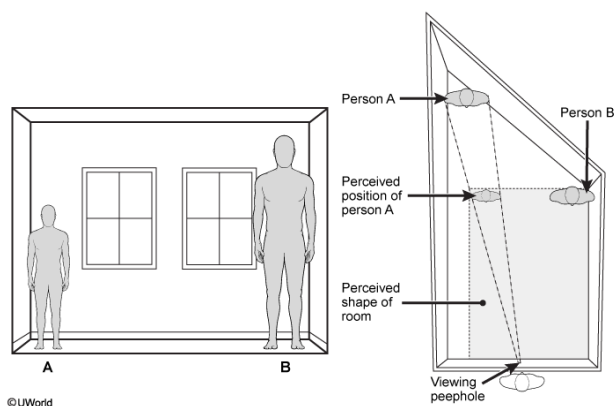
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perceptual distortions when landing on unfamiliar runways. During their training, pilots develop a mental image of how a typical runway appears as they descend and land. Later, they compare this prototypical shape to the runway they see before them to make adjustments in the slope of their descent. However, if a runway differs from what the pilot is used to (eg, in width or slope), it will differ from the pilot's mental image for a typical runway at that same altitude. For example, an unusually wide runway will appear to be closer than it actually is, and a narrow runway will seem to be further away. This can cause the pilot to approach at a dangerously steep or shallow slope.

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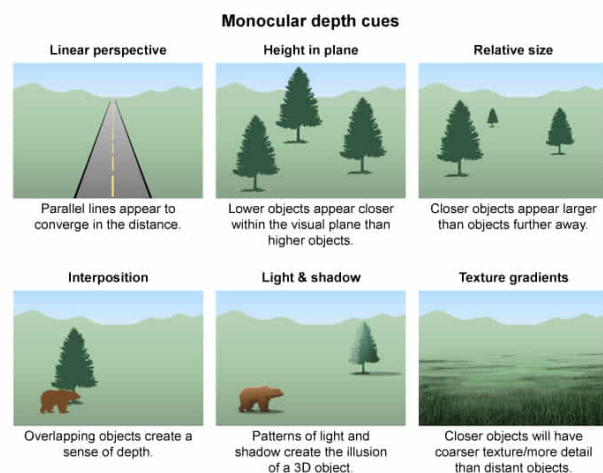
Pilots who experienced perceptual distortions while landing on an unusually wide runway were most likely affected by which type of monocular depth cue?

- ☐ A. Interposition [19%]
- ✓ ☒ B. Linear perspective [75%]
- ☐ C. Light and shadow [3%]
- ☐ D. Texture gradient [1%]

Correct

75% answered correctly

## Explanation:



Both monocular and binocular cues are used to determine depth and distance.

**Monocular cues** are depth cues that can be perceived using **one eye only** and often rely on **comparisons between objects**. Examples of monocular cues include:

- Interposition: Closer objects will overlap in front of objects that are further away (**Choice A**).
- Light and shadow: Highlights and shadows help to give objects the appearance of depth (**Choice C**).
- Texture gradient: Closer objects show more fine detail than objects that are further away (**Choice D**).
- Relative size: Further objects appear smaller than closer objects of the same size.
- Relative height: Further objects are perceived at higher distances within a field.
- Linear perspective: Distances between parallel lines appear narrower as they become further away.

A pilot will rely on visual cues from the runway to help determine her distance from it. As the aircraft moves closer to the runway, it takes up a larger portion of her visual field, and the relative distance between the nearest parallel edges of the runway will grow (ie, the linear perspective of the runway shifts as the pilot changes altitude). If the runway is much wider than runways with which the pilot is familiar, she is more likely to misperceive the distance of the unfamiliar runway because it will look different from a familiar runway at the same altitude.

### **Educational objective:**

Monocular cues are depth cues that can be perceived using one eye only and often rely on comparisons between objects. Examples include relative size, texture gradient changes, relative height, and the distance between parallel lines. Deviations from standard objects could interfere with monocular distance cues.

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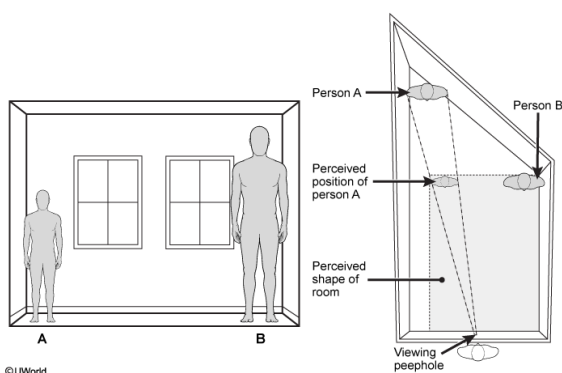
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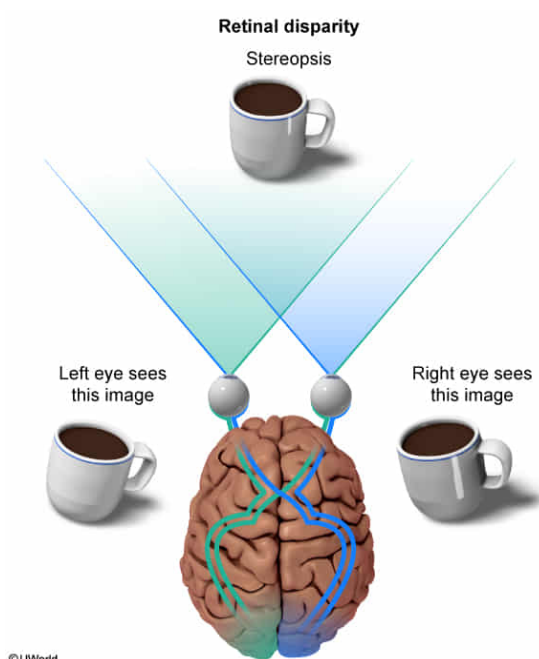
How does stereopsis contribute to the processing of two-dimensional retinal images into objects having three-dimensional depth?

- ☐ A. The presence of rods and cones allows for processing different types of light. [6%]
- ☐ B. Component attributes, such as color, motion, and form, are integrated in the occipital lobe. [28%]
- ✓ ☒ C. The relative location of each retina allows for different images of the object to be processed. [43%]
- ☐ D. The movement of ciliary muscles controlling the lens provides depth cues to the cortex. [20%]

Correct

43% answered correctly

### Explanation:



**Stereopsis** is the perception of depth that arises from the integrated information received from both eyes. **Binocular depth cues** allow accurate perception of depth through the integration of slightly different information from the left and right eyes. Binocular depth cues

include retinal disparity and convergence:

- **Retinal disparity**, or distance between the two eyes, results in slightly **different images** projected onto the respective retinas. The closer an object or scene is to the two eyes, the more dramatic the disparity. Higher-order processing in the brain compares the difference between the two-dimensional retinal images to help give rise to the perception of three-dimensional depth.
- **Convergence** occurs because of the **angular positioning of the eyes**: The closer an object is to the observer, the more the eyes turn inward (or converge) to focus on that object. The brain interprets the angle of convergence as an indication of distance from the observer.

Binocular cues are less informative at great distances because retinal disparity and convergence decrease as distance increases. Therefore, monocular cues become more important for depth perception over great distances.

**(Choice A)** Rods (light sensitive) and cones (color sensitive) are two different types of photoreceptors in the eye. Although their differences allow for better night vision and color vision, they do not contribute to depth perception.

**(Choice B)** Feature detection involves the analysis of component attributes or features of perceived objects, such as color, form, motion, and depth; feature detection does not explain *how* depth or three-dimensional form is interpreted.

**(Choice D)** As an object approaches the eye, ciliary muscles alter the shape of the lens. This is referred to as **accommodation** and is processed by the brain as a depth perception cue. However, accommodation is a monocular cue because it can be performed by only one eye and therefore does not contribute to stereopsis.

### **Educational objective:**

Binocular depth cues result in depth perception (stereopsis) based on information from both eyes. Retinal disparity allows for integration of slightly different images from each retina. Convergence refers to the angular positioning of the eyes as they focus on objects.

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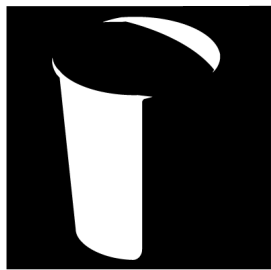


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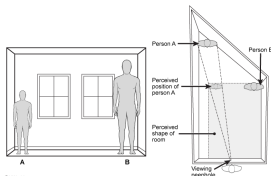
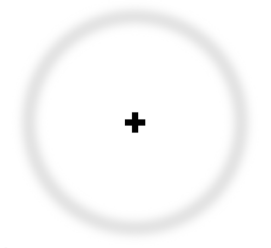


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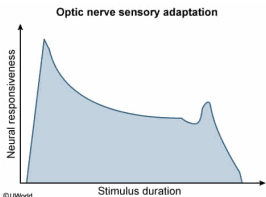


If you fixate on the black plus sign in the middle of this image without moving your eyes, the grey circle around the plus sign eventually disappears. Known as the Troxler effect, this occurs because of:

- ☐ A. opponent processing. [15%]
- ☐ B. interposition. [8%]
- ☒ C. sensory adaptation. [61%]
- ☐ D. the blind spot. [13%]

Correct  
61% answered correctly

Explanation:



**Sensory adaptation** is a diminished response by the nervous system when presented with an unchanging stimulus over a period of time. Decreased nervous system responsiveness results in the diminished perception of that stimulus.

In the above example, fixation on the plus sign results in an **unchanging** peripheral **visual stimulus**, which results in a **diminished response** by the optic nerve, and the circle seems to disappear. Even slight movement of the eye away from the fixation point will alter the stimulation of these nerves, thereby increasing their responsiveness to the changing stimulus and causing the circle to reappear.

Sensory adaptation can occur with any sensory system. For example, when one first walks into a kitchen where cookies are baking, the scent seems strong, but it fades as one continues to be exposed to it. Sensory adaptation also includes dark adaptation and light adaptation, or adaptation of vision with reduced and increased light levels, respectively.

(Choice A) Opponent-process theory states that color vision occurs because of the opposing responses of three sensory receptor complexes (red-green, blue-yellow, and black-white).

(Choice B) Interposition is a monocular depth cue that refers to one object partially blocking another (the fully visible object is perceived as being closer).

(Choice D) The blind spot is a small region between the central and peripheral visual fields where there are no photoreceptors because the optic nerve exits the retina at this location. The gradual disappearance of a visual stimulus, as occurs with the Troxler effect, cannot be explained by the blind spot.

**Educational objective:**

Sensory adaptation occurs when the neural response to an unchanging stimulus is dampened and the perception of that stimulus either disappears or is diminished. Sensory adaptation can occur with vision, hearing, smell, taste, and touch.

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